

HPC-VS Connectivity and Past-Year 5+ Drinking Frequency

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HPC-VS Binge Drinking Analysis

This R Markdown file runs the analyses and creates the figures reported in the manuscript and supplement:

1. Primary proportional-odds ordinal logistic models.
2. Permutation likelihood-ratio checks with 10,000 permutations.
3. DFBETAS influence sensitivity checks.
4. Sex-moderation models and conditional slopes.
5. Age sensitivity analyses.
6. Three-level alcohol-diagnosis follow-ups: no lifetime DSM-IV alcohol diagnosis, lifetime DSM-IV alcohol abuse, and lifetime DSM-IV alcohol dependence.
7. Diagnosis-group ANCOVAs for standardized left and right HPC-VS connectivity.
8. Manuscript Figures 2-5.

- `set.seed(23)`
- `N_PERM <- 10000`
- DFBETAS screens include `max |DFBETAS| > .20` and `2/sqrt(N)`

```
options(contrasts = c("contr.sum", "contr.poly"))
set.seed(23)

suppressPackageStartupMessages({
  library(tidyverse)
  library(MASS)
  library(rio)
  library(emmeans)
})

DATA_PATH <- "Data/averaged_data.csv"
PERM_SEED <- 23
N_PERM <- 10000
RMS_DFBETAS_CUTOFF <- 0.20

args <- commandArgs(trailingOnly = FALSE)
file_arg <- args[grepl("^--file=", args)]
SCRIPT_DIR <- if (length(file_arg) == 1) dirname(normalizePath(sub("^--file=", "", file_arg))) else getwd()
OUTPUT_DIR <- file.path(SCRIPT_DIR, "HPC_Ordinal_ManuscriptOrdered_Permutation_DFBETAS_Output")
dir.create(OUTPUT_DIR, recursive = TRUE, showWarnings = FALSE)

fmt_num <- function(x, digits = 3) {
```

```

    ifelse(is.na(x), "NA", formatC(x, format = "f", digits = digits))
  }

fmt_p <- function(p) {
  ifelse(
    is.na(p),
    "NA",
    ifelse(p < 0.001, "< .001", sub("^0", "", formatC(p, format = "f", digits = 3)))
  )
}

cat_line <- function(...) cat(..., "\n", sep = "")

fit_polr_safe <- function(formula, data) {
  suppressWarnings(
    MASS::polr(formula, data = data, Hess = TRUE, method = "logistic", model = TRUE)
  )
}

coef_table <- function(model) {
  ct <- coef(summary(model))
  p <- 2 * pnorm(abs(ct[, "t value"]), lower.tail = FALSE)

  tibble(
    term = rownames(ct),
    b = ct[, "Value"],
    SE = ct[, "Std. Error"],
    z = ct[, "t value"],
    p = p,
    OR = exp(ct[, "Value"])
  )
}

lr_stat <- function(reduced_model, full_model) {
  2 * (as.numeric(logLik(full_model)) - as.numeric(logLik(reduced_model)))
}

lr_test <- function(reduced_model, full_model, label) {
  lr <- lr_stat(reduced_model, full_model)
  df <- attr(logLik(full_model), "df") - attr(logLik(reduced_model), "df")
  tibble(
    test = label,
    df = df,
    LR = lr,
    p = pchisq(lr, df = df, lower.tail = FALSE)
  )
}

lm_block_test <- function(reduced_model, full_model, label) {
  a <- anova(reduced_model, full_model)
  tibble(
    test = label,
    df_num = a$Df[2],

```

```

    df_den = a$Res.Df[2],
    F = a$F[2],
    p = a$`Pr(>F)`[2]
  )
}

lm_coef_table <- function(model) {
  ct <- coef(summary(model))
  tibble(
    term = rownames(ct),
    b = ct[, "Estimate"],
    SE = ct[, "Std. Error"],
    t = ct[, "t value"],
    p = ct[, "Pr(>|t|)"]
  )
}

fmt_count_pct <- function(n, denom) {
  paste0(n, " (", fmt_num(100 * n / denom, 1), "%)")
}

perm_lr_test <- function(reduced_formula, full_formula, data, label, n_perm = N_PERM, seed = PERM_SEED) {
  set.seed(seed)

  reduced_obs <- fit_polr_safe(reduced_formula, data)
  full_obs <- fit_polr_safe(full_formula, data)
  observed_lr <- lr_stat(reduced_obs, full_obs)

  perm_stats <- replicate(n_perm, {
    dat_p <- data
    dat_p$y <- sample(dat_p$y, replace = FALSE)

    val <- try({
      red_p <- fit_polr_safe(reduced_formula, dat_p)
      full_p <- fit_polr_safe(full_formula, dat_p)
      lr_stat(red_p, full_p)
    }, silent = TRUE)

    if (inherits(val, "try-error") || !is.finite(val)) NA_real_ else val
  })

  perm_stats <- perm_stats[is.finite(perm_stats)]
  p_perm <- (sum(perm_stats >= observed_lr) + 1) / (length(perm_stats) + 1)

  tibble(
    test = label,
    LR = observed_lr,
    p_perm = p_perm,
    n_success = length(perm_stats)
  )
}

print_coef_line <- function(model, term, label = term) {

```

```

x <- coef_table(model) %>% filter(.data$term == !!term)
cat_line(
  " ", label,
  ": b = ", fmt_num(x$b, 4),
  ", SE = ", fmt_num(x$SE, 4),
  ", z = ", fmt_num(x$z, 3),
  ", p = ", fmt_p(x$p),
  ", OR = ", fmt_num(x$OR, 3)
)
}

print_coef_line_any <- function(model, terms, label = terms[1]) {
  x <- coef_table(model) %>% filter(.data$term %in% terms) %>% slice(1)
  if (nrow(x) == 0) {
    cat_line(" ", label, ": term not found. Available terms: ", paste(coef_table(model)$term, collapse = ", "))
  } else {
    cat_line(
      " ", label,
      ": b = ", fmt_num(x$b, 4),
      ", SE = ", fmt_num(x$SE, 4),
      ", z = ", fmt_num(x$z, 3),
      ", p = ", fmt_p(x$p),
      ", OR = ", fmt_num(x$OR, 3)
    )
  }
}

print_lr_line <- function(test_row) {
  cat_line(
    " ", test_row$test,
    ": LR chi-square(", test_row$df, ") = ", fmt_num(test_row$LR, 3),
    ", p = ", fmt_p(test_row$p)
  )
}

print_lm_block_line <- function(test_row) {
  cat_line(
    " ", test_row$test,
    ": F(", test_row$df_num, ", ", test_row$df_den, ") = ", fmt_num(test_row$F, 3),
    ", p = ", fmt_p(test_row$p)
  )
}

print_perm_line <- function(test_row) {
  cat_line(
    " ", test_row$test,
    ": observed LR = ", fmt_num(test_row$LR, 3),
    ", permutation p = ", fmt_p(test_row$p_perm),
    " (", test_row$n_success, " successful permutations)"
  )
}

dfbetas_screen <- function(formula, data, label) {

```

```

full_model <- fit_polr_safe(formula, data)
full_beta <- coef(full_model)
se_lookup <- coef_table(full_model) %>%
  filter(.data$term %in% names(full_beta)) %>%
  dplyr::select(term, SE)

purrr::map_dfr(seq_len(nrow(data)), function(i) {
  deleted_model <- fit_polr_safe(formula, droplevels(data[-i, , drop = FALSE]))
  deleted_beta <- coef(deleted_model)[names(full_beta)]

  tibble(
    model = label,
    row_id = data$row_id[i],
    Subject = data$Subject[i],
    deleted_position = i,
    term = names(full_beta),
    dfbetas = as.numeric((deleted_beta - full_beta) / se_lookup$SE)
  )
}) %>%
  mutate(abs_dfbetas = abs(dfbetas))
}

summarize_dfbetas_cases <- function(screen_long, cutoff, cutoff_label) {
  screen_long %>%
    mutate(flagged_term = abs_dfbetas > cutoff) %>%
    group_by(model, row_id, Subject) %>%
    summarize(
      cutoff_label = cutoff_label,
      cutoff = cutoff,
      max_abs_dfbetas = max(abs_dfbetas, na.rm = TRUE),
      flagged_any = any(flagged_term),
      flagged_terms = paste(term[flagged_term], collapse = "; "),
      .groups = "drop"
    ) %>%
    arrange(desc(max_abs_dfbetas))
}

dfbetas_refit <- function(label, data, reduced_formula, full_formula, target_term, screen_long, cutoff,
  cases <- summarize_dfbetas_cases(screen_long, cutoff, cutoff_label)
  flagged <- cases %>% filter(flagged_any)
  filtered_data <- data %>% filter(!row_id %in% flagged$row_id) %>% droplevels()

  reduced_full_sample <- fit_polr_safe(reduced_formula, data)
  full_full_sample <- fit_polr_safe(full_formula, data)
  full_coef <- coef_table(full_full_sample) %>% filter(.data$term == !!target_term)
  full_lr <- lr_test(reduced_full_sample, full_full_sample, "Full sample")

  reduced_filtered <- fit_polr_safe(reduced_formula, filtered_data)
  full_filtered <- fit_polr_safe(full_formula, filtered_data)
  filtered_coef <- coef_table(full_filtered) %>% filter(.data$term == !!target_term)
  filtered_lr <- lr_test(reduced_filtered, full_filtered, cutoff_label)

  result <- tibble(

```

```

    model = label,
    cutoff_label = cutoff_label,
    cutoff = cutoff,
    removed_n = nrow(flagged),
    retained_n = nrow(filtered_data),
    target_term = target_term,
    b = filtered_coef$b,
    SE = filtered_coef$SE,
    z = filtered_coef$z,
    p = filtered_coef$p,
    OR = filtered_coef$OR,
    LR = filtered_lr$LR,
    LR_df = filtered_lr$df,
    LR_p = filtered_lr$p
  )

  list(
    cases = cases,
    flagged = flagged,
    filtered_data = filtered_data,
    filtered_model = full_filtered,
    result = result,
    full_result = tibble(
      model = label,
      cutoff_label = "Full sample",
      cutoff = NA_real_,
      removed_n = 0L,
      retained_n = nrow(data),
      target_term = target_term,
      b = full_coef$b,
      SE = full_coef$SE,
      z = full_coef$z,
      p = full_coef$p,
      OR = full_coef$OR,
      LR = full_lr$LR,
      LR_df = full_lr$df,
      LR_p = full_lr$p
    )
  )
}

run_dfbetas_checks <- function(label, data, reduced_formula, full_formula, target_term) {
  bkw_cutoff <- 2 / sqrt(nrow(data))
  screen_long <- dfbetas_screen(full_formula, data, label)

  rms <- dfbetas_refit(
    label = label,
    data = data,
    reduced_formula = reduced_formula,
    full_formula = full_formula,
    target_term = target_term,
    screen_long = screen_long,
    cutoff = RMS_DFBETAS_CUTOFF,

```

```

    cutoff_label = "max |DFBETAS| > .20"
  )

  bkw <- dfbetas_refit(
    label = label,
    data = data,
    reduced_formula = reduced_formula,
    full_formula = full_formula,
    target_term = target_term,
    screen_long = screen_long,
    cutoff = bkw_cutoff,
    cutoff_label = paste0("max |DFBETAS| > 2/sqrt(N) = ", fmt_num(bkw_cutoff, 4))
  )

  result <- bind_rows(rms$full_result, rms$result, bkw$result)

  readr::write_csv(screen_long, file.path(OUTPUT_DIR, paste0(label, "_dfbetas_long.csv")))
  readr::write_csv(rms$cases, file.path(OUTPUT_DIR, paste0(label, "_dfbetas_cases_rms020.csv")))
  readr::write_csv(bkw$cases, file.path(OUTPUT_DIR, paste0(label, "_dfbetas_cases_bkw.csv")))

  cat_line(" DFBETAS sensitivity for ", label, ":")
  for (i in seq_len(nrow(result))) {
    x <- result[i, ]
    cat_line(
      "      ", x$cutoff_label,
      ": removed N = ", x$removed_n,
      ", retained N = ", x$retained_n,
      ", b = ", fmt_num(x$b, 4),
      ", SE = ", fmt_num(x$SE, 4),
      ", z = ", fmt_num(x$z, 3),
      ", p = ", fmt_p(x$p),
      ", LR chi-square(", x$LR_df, ") = ", fmt_num(x$LR, 3),
      ", LR p = ", fmt_p(x$LR_p)
    )
  }

  list(screen_long = screen_long, rms = rms, bkw = bkw, result = result)
}

simple_slopes <- function(model, var_name, label) {
  trends <- emmeans::emtrends(model, specs = ~ Sex, var = var_name, mode = "latent")
  slopes <- as.data.frame(summary(trends, infer = c(TRUE, TRUE))) %>%
    as_tibble() %>%
    transmute(
      analysis = label,
      contrast = as.character(Sex),
      b = !!sym(paste0(var_name, ".trend")),
      SE = SE,
      z = z.ratio,
      p = p.value,
      lower.CL = asymp.LCL,
      upper.CL = asymp.UCL
    )
}

```

```

diff <- as.data.frame(summary(pairs(trends), infer = c(TRUE, TRUE))) %>%
  as_tibble() %>%
  transmute(
    analysis = label,
    contrast = as.character(contrast),
    b = estimate,
    SE = SE,
    z = z.ratio,
    p = p.value,
    lower.CL = asymp.LCL,
    upper.CL = asymp.UCL
  )

bind_rows(slopes, diff)
}

print_simple_slopes <- function(slopes_tbl, label) {
  cat_line(" Simple slopes for ", label, ":")
  for (i in seq_len(nrow(slopes_tbl))) {
    x <- slopes_tbl[i, ]
    cat_line(
      " ", x$contrast,
      ": b = ", fmt_num(x$b, 4),
      ", SE = ", fmt_num(x$SE, 4),
      ", z = ", fmt_num(x$z, 3),
      ", p = ", fmt_p(x$p),
      ", 95% CI [", fmt_num(x$lower.CL, 4), ", ", fmt_num(x$upper.CL, 4), "]"
    )
  }
}

HCP_dat <- rio::import(DATA_PATH)

analysis_dat <- HCP_dat %>%
  mutate(
    Subject = as.character(Subject),
    Sex = factor(trimws(Gender), levels = c("F", "M")),
    binge_freq_rc = case_when(
      SSAGA_Alc_12_Frq_5plus == 5 ~ 1L,
      SSAGA_Alc_12_Frq_5plus == 4 ~ 2L,
      SSAGA_Alc_12_Frq_5plus == 3 ~ 3L,
      SSAGA_Alc_12_Frq_5plus %in% c(1L, 2L) ~ 4L,
      TRUE ~ NA_integer_
    ),
    binge_freq_rc_f = factor(
      binge_freq_rc,
      levels = c(1, 2, 3, 4),
      labels = c("Never", "1-11 d/year", "1-3 d/month", "Weekly or greater"),
      ordered = TRUE
    ),
    alcohol_abuse_dx = case_when(
      SSAGA_Alc_D4_Ab_Dx == 5 ~ TRUE,
      SSAGA_Alc_D4_Ab_Dx == 1 ~ FALSE,

```

```

    TRUE ~ NA
  ),
  alcohol_dependence_dx = case_when(
    SSAGA_Alc_D4_Dp_Dx == 5 ~ TRUE,
    SSAGA_Alc_D4_Dp_Dx == 1 ~ FALSE,
    TRUE ~ NA
  ),
  AUD = factor(
    if_else(alcohol_abuse_dx | alcohol_dependence_dx, "AUD", "No AUD"),
    levels = c("No AUD", "AUD")
  ),
  DxGroup = factor(
    case_when(
      alcohol_dependence_dx ~ "Dependence",
      alcohol_abuse_dx ~ "Abuse",
      alcohol_abuse_dx == FALSE & alcohol_dependence_dx == FALSE ~ "No AUD",
      TRUE ~ NA_character_
    ),
    levels = c("No AUD", "Abuse", "Dependence")
  )
) %>%
transmute(
  row_id = row_number(),
  Subject,
  Sex,
  y = binge_freq_rc_f,
  alcohol_abuse_dx,
  alcohol_dependence_dx,
  AUD,
  DxGroup,
  left_hpc_vs = as.numeric(limbic_Left),
  right_hpc_vs = as.numeric(limbic_Right),
  ICV = as.numeric(FS_IntraCranial_Vol),
  Age = as.numeric(Age_in_Yrs)
) %>%
filter(
  is.finite(left_hpc_vs),
  is.finite(right_hpc_vs),
  is.finite(ICV),
  is.finite(Age),
  !is.na(y),
  !is.na(Sex),
  !is.na(AUD),
  !is.na(DxGroup)
) %>%
mutate(
  c_left_hpc_vs = as.numeric(scale(left_hpc_vs)),
  c_right_hpc_vs = as.numeric(scale(right_hpc_vs)),
  c_ICV = as.numeric(scale(ICV)),
  c_Age = as.numeric(scale(Age))
) %>%
droplevels()

```

```

contrasts(analysis_dat$AUD) <- matrix(
  c(0, 1),
  ncol = 1,
  dimnames = list(levels(analysis_dat$AUD), "AUD")
)
contrasts(analysis_dat$DxGroup) <- matrix(
  c(0, 1, 0, 0, 0, 1),
  ncol = 2,
  dimnames = list(levels(analysis_dat$DxGroup), c("Abuse", "Dependence"))
)

left_dat <- analysis_dat %>%
  dplyr::select(row_id, Subject, Sex, y, c_left_hpc_vs, c_ICV, c_Age) %>%
  droplevels()

right_dat <- analysis_dat %>%
  dplyr::select(row_id, Subject, Sex, y, c_right_hpc_vs, c_ICV, c_Age) %>%
  droplevels()

cat_line(strrep("=", 90))

```

```

cat_line("HPC ORDINAL WORKFLOW: PRIMARY MODELS FOLLOWED BY PERMUTATION AND DFBETAS CHECKS")

```

HPC ORDINAL WORKFLOW: PRIMARY MODELS FOLLOWED BY PERMUTATION AND DFBETAS CHECKS

```

cat_line(strrep("=", 90))

```

```

cat_line("Analysis N = ", nrow(analysis_dat))

```

Analysis N = 593

```

cat_line("Permutation settings: seed = ", PERM_SEED, "; target permutations = ", N_PERM, ".")

```

Permutation settings: seed = 23; target permutations = 10000.

```

cat_line("Permutation method: shuffle ordinal outcome labels, refit reduced/full proportional-odds models")

```

Permutation method: shuffle ordinal outcome labels, refit reduced/full proportional-odds models, and compare observed LR to empirical null LR distribution.

```

cat_line("Permutation p-values use the plus-one correction: (number of permuted LR values >= observed LR + 1) / (successful permutations + 1).")

```

Permutation p-values use the plus-one correction: $(\text{number of permuted LR values} \geq \text{observed LR} + 1) / (\text{successful permutations} + 1)$.

```
cat_line("DFBETAS method: refit the proportional-odds model after deleting one participant at a time; DFBETAS
```

DFBETAS method: refit the proportional-odds model after deleting one participant at a time; DFBETAS
= leave-one-out coefficient change / full-sample SE.

```
cat_line("DFBETAS screens: rms-style max |DFBETAS| > .20 and Belsley-Kuh-Welsch max |DFBETAS| > 2/sqrt(N)
```

DFBETAS screens: rms-style max |DFBETAS| > .20 and Belsley-Kuh-Welsch max |DFBETAS| > 2/sqrt(N).

```
left_base <- fit_polr_safe(y ~ c_ICV, data = left_dat)
left_main <- fit_polr_safe(y ~ c_ICV + c_left_hpc_vs, data = left_dat)
right_base <- fit_polr_safe(y ~ c_ICV, data = right_dat)
right_main <- fit_polr_safe(y ~ c_ICV + c_right_hpc_vs, data = right_dat)

cat_line("\n", strrep("=", 90))
```

```
cat_line("SECTION 1: MAIN-EFFECT MODELS")
```

SECTION 1: MAIN-EFFECT MODELS

```
cat_line(strrep("=", 90))
```

```
cat_line("\n1A. Left hemisphere main effect")
```

1A. Left hemisphere main effect

```
print_coef_line(left_main, "c_left_hpc_vs", "Left HPC-VS connectivity")
```

Left HPC-VS connectivity: b = 0.1708, SE = 0.0770, z = 2.219, p = .027, OR = 1.186

```
left_main_lr <- lr_test(left_base, left_main, "Left main effect: add HPC-VS connectivity")
print_lr_line(left_main_lr)
```

Left main effect: add HPC-VS connectivity: LR chi-square(1) = 4.946, p = .026

```
cat_line("\nChecks for 1A")
```

Checks for 1A

```
left_main_perm <- perm_lr_test(
  y ~ c_ICV,
  y ~ c_ICV + c_left_hpc_vs,
  left_dat,
  "Permutation LR, left main effect"
)
print_perm_line(left_main_perm)
```

Permutation LR, left main effect: observed LR = 4.946, permutation p = .025 (10000 successful permutations)

```
left_main_dfb <- run_dfbetas_checks(  
  "left_main",  
  left_dat,  
  y ~ c_ICV,  
  y ~ c_ICV + c_left_hpc_vs,  
  "c_left_hpc_vs"  
)
```

DFBETAS sensitivity for left_main: Full sample: removed N = 0, retained N = 593, b = 0.1708, SE = 0.0770, z = 2.219, p = .027, LR chi-square(1) = 4.946, LR p = .026 max |DFBETAS| > .20: removed N = 4, retained N = 589, b = 0.1837, SE = 0.0780, z = 2.356, p = .018, LR chi-square(1) = 5.571, LR p = .018 max |DFBETAS| > 2/sqrt(N) = 0.0821: removed N = 61, retained N = 532, b = 0.2404, SE = 0.0899, z = 2.674, p = .007, LR chi-square(1) = 7.201, LR p = .007

```
cat_line("\n1B. Right hemisphere main effect")
```

1B. Right hemisphere main effect

```
print_coef_line(right_main, "c_right_hpc_vs", "Right HPC-VS connectivity")
```

Right HPC-VS connectivity: b = 0.0219, SE = 0.0762, z = 0.287, p = .774, OR = 1.022

```
right_main_lr <- lr_test(right_base, right_main, "Right main effect: add HPC-VS connectivity")  
print_lr_line(right_main_lr)
```

Right main effect: add HPC-VS connectivity: LR chi-square(1) = 0.082, p = .774

```
cat_line("\nChecks for 1B")
```

Checks for 1B

```
right_main_perm <- perm_lr_test(  
  y ~ c_ICV,  
  y ~ c_ICV + c_right_hpc_vs,  
  right_dat,  
  "Permutation LR, right main effect"  
)  
print_perm_line(right_main_perm)
```

Permutation LR, right main effect: observed LR = 0.082, permutation p = .774 (10000 successful permutations)

```
right_main_dfb <- run_dfbetas_checks(  
  "right_main",  
  right_dat,  
  y ~ c_ICV,  
  y ~ c_ICV + c_right_hpc_vs,  
  "c_right_hpc_vs"  
)
```

DFBETAS sensitivity for right_main: Full sample: removed N = 0, retained N = 593, b = 0.0219, SE = 0.0762, z = 0.287, p = .774, LR chi-square(1) = 0.082, LR p = .774 max |DFBETAS| > .20: removed N = 4, retained N = 589, b = 0.0361, SE = 0.0775, z = 0.466, p = .641, LR chi-square(1) = 0.217, LR p = .641 max |DFBETAS| > 2/sqrt(N) = 0.0821: removed N = 62, retained N = 531, b = 0.0086, SE = 0.0882, z = 0.097, p = .923, LR chi-square(1) = 0.009, LR p = .923

```
left_sex <- fit_polr_safe(y ~ c_ICV + c_left_hpc_vs + Sex, data = left_dat)
left_int <- fit_polr_safe(y ~ c_ICV + c_left_hpc_vs * Sex, data = left_dat)
right_sex <- fit_polr_safe(y ~ c_ICV + c_right_hpc_vs + Sex, data = right_dat)
right_int <- fit_polr_safe(y ~ c_ICV + c_right_hpc_vs * Sex, data = right_dat)

cat_line("\n", strrep("=", 90))
```

```
cat_line("SECTION 2: SEX-MODERATION MODELS")
```

SECTION 2: SEX-MODERATION MODELS

```
cat_line(strrep("=", 90))
```

```
cat_line("\n2A. Left hemisphere sex-moderation model")
```

2A. Left hemisphere sex-moderation model

```
print_coef_line(left_int, "c_left_hpc_vs:Sex1", "Left HPC-VS connectivity x Sex")
```

Left HPC-VS connectivity x Sex: b = 0.0150, SE = 0.0776, z = 0.194, p = .846, OR = 1.015

```
left_sex_lr <- lr_test(left_main, left_sex, "Left model: add sex")
left_int_lr <- lr_test(left_sex, left_int, "Left model: add HPC-VS connectivity x Sex")
print_lr_line(left_sex_lr)
```

Left model: add sex: LR chi-square(1) = 14.143, p = < .001

```
print_lr_line(left_int_lr)
```

Left model: add HPC-VS connectivity x Sex: LR chi-square(1) = 0.038, p = .846

```
cat_line("\nChecks for 2A")
```

Checks for 2A

```
left_sex_perm <- perm_lr_test(
  y ~ c_ICV + c_left_hpc_vs,
  y ~ c_ICV + c_left_hpc_vs + Sex,
  left_dat,
  "Permutation LR, left model add sex")
```

```
)
left_int_perm <- perm_lr_test(
  y ~ c_ICV + c_left_hpc_vs + Sex,
  y ~ c_ICV + c_left_hpc_vs * Sex,
  left_dat,
  "Permutation LR, left interaction"
)
print_perm_line(left_sex_perm)
```

Permutation LR, left model add sex: observed LR = 14.143, permutation p = < .001 (10000 successful permutations)

```
print_perm_line(left_int_perm)
```

Permutation LR, left interaction: observed LR = 0.038, permutation p = .851 (10000 successful permutations)

```
left_int_dfb <- run_dfbetas_checks(
  "left_interaction",
  left_dat,
  y ~ c_ICV + c_left_hpc_vs + Sex,
  y ~ c_ICV + c_left_hpc_vs * Sex,
  "c_left_hpc_vs:Sex1"
)
```

DFBETAS sensitivity for left_interaction: Full sample: removed N = 0, retained N = 593, b = 0.0150, SE = 0.0776, z = 0.194, p = .846, LR chi-square(1) = 0.038, LR p = .846 max |DFBETAS| > .20: removed N = 3, retained N = 590, b = 0.0003, SE = 0.0786, z = 0.003, p = .997, LR chi-square(1) = 0.000, LR p = .997 max |DFBETAS| > 2/sqrt(N) = 0.0821: removed N = 104, retained N = 489, b = -0.0028, SE = 0.0967, z = -0.029, p = .977, LR chi-square(1) = 0.001, LR p = .977

```
cat_line("\nLeft conditional slopes by sex")
```

Left conditional slopes by sex

```
left_slopes_full <- simple_slopes(left_int, "c_left_hpc_vs", "Full sample")
left_slopes_rms <- simple_slopes(left_int_dfb$rms$filtered_model, "c_left_hpc_vs", "After max |DFBETAS|
left_slopes_bkw <- simple_slopes(left_int_dfb$bkw$filtered_model, "c_left_hpc_vs", "After max |DFBETAS|
print_simple_slopes(left_slopes_full, "left interaction, full sample")
```

Simple slopes for left interaction, full sample: F: b = 0.1681, SE = 0.1023, z = 1.644, p = .100, 95% CI [-0.0324, 0.3686] M: b = 0.1380, SE = 0.1169, z = 1.181, p = .238, 95% CI [-0.0911, 0.3671] F - M: b = 0.0301, SE = 0.1552, z = 0.194, p = .846, 95% CI [-0.2741, 0.3343]

```
print_simple_slopes(left_slopes_rms, "left interaction after .20 DFBETAS removal")
```

Simple slopes for left interaction after .20 DFBETAS removal: F: b = 0.1645, SE = 0.1025, z = 1.604, p = .109, 95% CI [-0.0365, 0.3655] M: b = 0.1639, SE = 0.1193, z = 1.374, p = .169, 95% CI [-0.0698, 0.3977] F - M: b = 0.0005, SE = 0.1572, z = 0.003, p = .997, 95% CI [-0.3075, 0.3086]

```
print_simple_slopes(left_slopes_bkw, "left interaction after 2/sqrt(N) DFBETAS removal")
```

Simple slopes for left interaction after 2/sqrt(N) DFBETAS removal: F: $b = 0.2331$, $SE = 0.1149$, $z = 2.028$, $p = .043$, 95% CI [0.0078, 0.4584] M: $b = 0.2386$, $SE = 0.1556$, $z = 1.534$, $p = .125$, 95% CI [-0.0663, 0.5436] F - M: $b = -0.0056$, $SE = 0.1934$, $z = -0.029$, $p = .977$, 95% CI [-0.3846, 0.3735]

```
cat_line("\n2B. Right hemisphere sex-moderation model")
```

2B. Right hemisphere sex-moderation model

```
print_coef_line(right_int, "c_right_hpc_vs:Sex1", "Right HPC-VS connectivity x Sex")
```

Right HPC-VS connectivity x Sex: $b = 0.1676$, $SE = 0.0777$, $z = 2.156$, $p = .031$, OR = 1.182

```
right_sex_lr <- lr_test(right_main, right_sex, "Right model: add sex")
right_int_lr <- lr_test(right_sex, right_int, "Right model: add HPC-VS connectivity x Sex")
print_lr_line(right_sex_lr)
```

Right model: add sex: LR chi-square(1) = 14.974, $p = < .001$

```
print_lr_line(right_int_lr)
```

Right model: add HPC-VS connectivity x Sex: LR chi-square(1) = 4.651, $p = .031$

```
cat_line("\nChecks for 2B")
```

Checks for 2B

```
right_sex_perm <- perm_lr_test(
  y ~ c_ICV + c_right_hpc_vs,
  y ~ c_ICV + c_right_hpc_vs + Sex,
  right_dat,
  "Permutation LR, right model add sex"
)
right_int_perm <- perm_lr_test(
  y ~ c_ICV + c_right_hpc_vs + Sex,
  y ~ c_ICV + c_right_hpc_vs * Sex,
  right_dat,
  "Permutation LR, right interaction"
)
print_perm_line(right_sex_perm)
```

Permutation LR, right model add sex: observed LR = 14.974, permutation $p = < .001$ (10000 successful permutations)

```
print_perm_line(right_int_perm)
```

Permutation LR, right interaction: observed LR = 4.651, permutation $p = .029$ (10000 successful permutations)

```
right_int_dfb <- run_dfbetas_checks(
  "right_interaction",
  right_dat,
  y ~ c_ICV + c_right_hpc_vs + Sex,
  y ~ c_ICV + c_right_hpc_vs * Sex,
  "c_right_hpc_vs:Sex1"
)
```

DFBETAS sensitivity for right_interaction: Full sample: removed N = 0, retained N = 593, b = 0.1676, SE = 0.0777, z = 2.156, p = .031, LR chi-square(1) = 4.651, LR p = .031 max |DFBETAS| > .20: removed N = 4, retained N = 589, b = 0.1780, SE = 0.0801, z = 2.223, p = .026, LR chi-square(1) = 4.960, LR p = .026 max |DFBETAS| > 2/sqrt(N) = 0.0821: removed N = 103, retained N = 490, b = 0.3054, SE = 0.0938, z = 3.257, p = .001, LR chi-square(1) = 10.697, LR p = .001

```
cat_line("\nRight conditional slopes by sex")
```

Right conditional slopes by sex

```
right_slopes_full <- simple_slopes(right_int, "c_right_hpc_vs", "Full sample")
right_slopes_rms <- simple_slopes(right_int_dfb$rms$filtered_model, "c_right_hpc_vs", "After max |DFBETAS| > .20 removal")
right_slopes_bkw <- simple_slopes(right_int_dfb$bkw$filtered_model, "c_right_hpc_vs", "After max |DFBETAS| > 2/sqrt(N) removal")
print_simple_slopes(right_slopes_full, "right interaction, full sample")
```

Simple slopes for right interaction, full sample: F: b = 0.1491, SE = 0.0988, z = 1.508, p = .132, 95% CI [-0.0447, 0.3428] M: b = -0.1860, SE = 0.1200, z = -1.551, p = .121, 95% CI [-0.4212, 0.0491] F - M: b = 0.3351, SE = 0.1554, z = 2.156, p = .031, 95% CI [0.0305, 0.6397]

```
print_simple_slopes(right_slopes_rms, "right interaction after .20 DFBETAS removal")
```

Simple slopes for right interaction after .20 DFBETAS removal: F: b = 0.1571, SE = 0.0992, z = 1.585, p = .113, 95% CI [-0.0372, 0.3515] M: b = -0.1990, SE = 0.1258, z = -1.582, p = .114, 95% CI [-0.4454, 0.0475] F - M: b = 0.3561, SE = 0.1602, z = 2.223, p = .026, 95% CI [0.0422, 0.6700]

```
print_simple_slopes(right_slopes_bkw, "right interaction after 2/sqrt(N) DFBETAS removal")
```

Simple slopes for right interaction after 2/sqrt(N) DFBETAS removal: F: b = 0.2843, SE = 0.1128, z = 2.521, p = .012, 95% CI [0.0633, 0.5054] M: b = -0.3265, SE = 0.1492, z = -2.188, p = .029, 95% CI [-0.6190, -0.0340] F - M: b = 0.6108, SE = 0.1875, z = 3.257, p = .001, 95% CI [0.2433, 0.9783]

```
cat_line("\n", strrep("=", 90))
```

```
cat_line("SECTION 3: AGE SENSITIVITY MODELS")
```

SECTION 3: AGE SENSITIVITY MODELS

```
cat_line(strrep("=", 90))
```

```
cat_line("\n3A. Left hemisphere main effect with age covariate")
```

3A. Left hemisphere main effect with age covariate

```
left_age_reduced <- fit_polr_safe(y ~ c_ICV + c_Age, data = left_dat)
left_age_full <- fit_polr_safe(y ~ c_ICV + c_Age + c_left_hpc_vs, data = left_dat)
print_coef_line(left_age_full, "c_left_hpc_vs", "Left HPC-VS connectivity, age-adjusted")
```

Left HPC-VS connectivity, age-adjusted: $b = 0.1557$, $SE = 0.0774$, $z = 2.013$, $p = .044$, $OR = 1.169$

```
left_age_lr <- lr_test(left_age_reduced, left_age_full, "Left + age: add HPC-VS connectivity")
print_lr_line(left_age_lr)
```

Left + age: add HPC-VS connectivity: LR chi-square(1) = 4.065, $p = .044$

```
cat_line("\n3B. Right hemisphere interaction with age covariate")
```

3B. Right hemisphere interaction with age covariate

```
right_age_reduced <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_right_hpc_vs + Sex,
  data = right_dat
)
right_age_full <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_right_hpc_vs * Sex,
  data = right_dat
)
print_coef_line(right_age_full, "c_right_hpc_vs:Sex1", "Right HPC-VS connectivity x Sex, age-adjusted")
```

Right HPC-VS connectivity x Sex, age-adjusted: $b = 0.1725$, $SE = 0.0778$, $z = 2.216$, $p = .027$, $OR = 1.188$

```
right_age_lr <- lr_test(right_age_reduced, right_age_full, "Right + age: add HPC-VS connectivity x Sex")
print_lr_line(right_age_lr)
```

Right + age: add HPC-VS connectivity x Sex: LR chi-square(1) = 4.914, $p = .027$

```
cat_line("\n3C. Left hemisphere continuous-age moderation")
```

3C. Left hemisphere continuous-age moderation

```
left_age_mod_reduced <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_left_hpc_vs,
  data = left_dat
)
left_age_mod_full <- fit_polr_safe(
  y ~ c_ICV + c_Age * c_left_hpc_vs,
  data = left_dat
)
print_coef_line(left_age_mod_full, "c_Age:c_left_hpc_vs", "Left HPC-VS connectivity x Age")
```

Left HPC-VS connectivity x Age: $b = 0.0215$, $SE = 0.0810$, $z = 0.265$, $p = .791$, $OR = 1.022$

```
left_age_mod_lr <- lr_test(left_age_mod_reduced, left_age_mod_full, "Left: add HPC-VS connectivity x Age")
print_lr_line(left_age_mod_lr)
```

Left: add HPC-VS connectivity x Age: LR chi-square(1) = 0.070, p = .791

```
cat_line("\n3D. Right hemisphere continuous-age moderation")
```

3D. Right hemisphere continuous-age moderation

```
right_age_mod_reduced <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_right_hpc_vs * Sex,
  data = right_dat
)
right_age_mod_full <- fit_polr_safe(
  y ~ c_ICV + c_Age + c_right_hpc_vs * Sex + c_right_hpc_vs:c_Age + Sex:c_Age + c_right_hpc_vs:Sex:c_Age,
  data = right_dat
)
print_coef_line_any(
  right_age_mod_full,
  c("c_right_hpc_vs:Sex1:c_Age", "c_Age:c_right_hpc_vs:Sex1", "c_right_hpc_vs:c_Age:Sex1"),
  "Right HPC-VS connectivity x Sex x Age"
)
```

Right HPC-VS connectivity x Sex x Age: b = -0.0309, SE = 0.0840, z = -0.367, p = .713, OR = 0.970

```
right_age_mod_lr <- lr_test(right_age_mod_reduced, right_age_mod_full, "Right: add age moderation terms")
print_lr_line(right_age_mod_lr)
```

Right: add age moderation terms: LR chi-square(3) = 5.169, p = .160

```
cat_line("\n", strrep("=", 90))
```

```
cat_line("SECTION 4: ALCOHOL DIAGNOSIS DESCRIPTIVES AND EXPLORATORY CHECKS")
```

SECTION 4: ALCOHOL DIAGNOSIS DESCRIPTIVES AND EXPLORATORY CHECKS

```
cat_line(strrep("=", 90))
```

```
cat_line("Diagnosis variables available in this dataset are DSM-IV alcohol abuse/dependence criteria-met")
```

Diagnosis variables available in this dataset are DSM-IV alcohol abuse/dependence criteria-met indicators.

```
sex_totals <- table(analysis_dat$Sex)
total_n <- nrow(analysis_dat)
female_n <- unname(sex_totals["F"])
```

```

male_n <- unname(sex_totals["M"])
female_abuse <- sum(analysis_dat$Sex == "F" & analysis_dat$alcohol_abuse_dx)
male_abuse <- sum(analysis_dat$Sex == "M" & analysis_dat$alcohol_abuse_dx)
total_abuse <- sum(analysis_dat$alcohol_abuse_dx)
female_dependence <- sum(analysis_dat$Sex == "F" & analysis_dat$alcohol_dependence_dx)
male_dependence <- sum(analysis_dat$Sex == "M" & analysis_dat$alcohol_dependence_dx)
total_dependence <- sum(analysis_dat$alcohol_dependence_dx)
female_aud <- sum(analysis_dat$Sex == "F" & analysis_dat$AUD == "AUD")
male_aud <- sum(analysis_dat$Sex == "M" & analysis_dat$AUD == "AUD")
total_aud <- sum(analysis_dat$AUD == "AUD")

diagnosis_counts <- tibble(
  Characteristic = c(
    "Sample size (n)",
    "DSM-IV lifetime alcohol diagnosis: alcohol abuse, n (%)",
    "DSM-IV lifetime alcohol diagnosis: alcohol dependence, n (%)",
    "Any lifetime DSM-IV alcohol use disorder, n (%)"
  ),
  Female = c(
    as.character(female_n),
    fmt_count_pct(female_abuse, female_n),
    fmt_count_pct(female_dependence, female_n),
    fmt_count_pct(female_aud, female_n)
  ),
  Male = c(
    as.character(male_n),
    fmt_count_pct(male_abuse, male_n),
    fmt_count_pct(male_dependence, male_n),
    fmt_count_pct(male_aud, male_n)
  ),
  Total = c(
    as.character(total_n),
    fmt_count_pct(total_abuse, total_n),
    fmt_count_pct(total_dependence, total_n),
    fmt_count_pct(total_aud, total_n)
  )
)

cat_line("\n4A. Lifetime alcohol disorder diagnosis table")

```

4A. Lifetime alcohol disorder diagnosis table

```
print(diagnosis_counts)
```

A tibble: 4 x 4

Characteristic	Female	Male	Total
1 Sample size (n)	332	261	593
2 DSM-IV lifetime alcohol diagnosis: alcohol abuse, n (%)	45 (1~ 59 (~ 104 ~ 3	DSM-IV lifetime alcohol	
diagnosis: alcohol dependence, n (~ 9 (2~ 15 (~ 24 (~ 4	Any lifetime DSM-IV alcohol use disorder, n (%)		
54 (1~ 74 (~ 128 ~			

```
cat_line("\nDiagnosis group counts for connectivity ANCOVAs")
```

Diagnosis group counts for connectivity ANCOVAs

```
print(table(analysis_dat$DxGroup))
```

```
No AUD      Abuse Dependence
  465        104         24
```

```
cat_line("\n4B. Three-level diagnosis-group moderation of ordinal drinking-frequency models")
```

4B. Three-level diagnosis-group moderation of ordinal drinking-frequency models

```
cat_line("  Diagnosis groups: No AUD, DSM-IV alcohol abuse, DSM-IV alcohol dependence.")
```

Diagnosis groups: No AUD, DSM-IV alcohol abuse, DSM-IV alcohol dependence.

```
left_dx_mod_reduced <- fit_polr_safe(y ~ c_ICV + DxGroup + c_left_hpc_vs, data = analysis_dat)
left_dx_mod_full <- fit_polr_safe(y ~ c_ICV + DxGroup * c_left_hpc_vs, data = analysis_dat)
left_dx_mod_lr <- lr_test(left_dx_mod_reduced, left_dx_mod_full, "Left: add diagnosis group x HPC-VS co
print_coef_line_any(
  left_dx_mod_full,
  c(
    "DxGroupAbuse:c_left_hpc_vs",
    "c_left_hpc_vs:DxGroupAbuse",
    "DxGroupDependence:c_left_hpc_vs",
    "c_left_hpc_vs:DxGroupDependence"
  ),
  "Three-level diagnosis group x Left HPC-VS connectivity"
)
```

Three-level diagnosis group x Left HPC-VS connectivity: $b = -0.0752$, $SE = 0.2006$, $z = -0.375$, $p = .708$, $OR = 0.928$

```
print_lr_line(left_dx_mod_lr)
```

Left: add diagnosis group x HPC-VS connectivity: $LR \text{ chi-square}(2) = 0.389$, $p = .823$

```
right_dx_mod_reduced <- fit_polr_safe(
  y ~ c_ICV + DxGroup + c_right_hpc_vs * Sex,
  data = analysis_dat
)
right_dx_mod_full <- fit_polr_safe(
  y ~ c_ICV + c_right_hpc_vs * Sex * DxGroup,
  data = analysis_dat
)
right_dx_mod_lr <- lr_test(right_dx_mod_reduced, right_dx_mod_full, "Right: add diagnosis-group moderat
print_coef_line_any(
  right_dx_mod_full,
  c(
```

```

    "c_right_hpc_vs:Sex1:DxGroupAbuse",
    "c_right_hpc_vs:DxGroupAbuse:Sex1",
    "Sex1:c_right_hpc_vs:DxGroupAbuse",
    "c_right_hpc_vs:Sex1:DxGroupDependence",
    "c_right_hpc_vs:DxGroupDependence:Sex1",
    "Sex1:c_right_hpc_vs:DxGroupDependence"
  ),
  "Right HPC-VS connectivity x Sex x three-level diagnosis group"
)

```

Right HPC-VS connectivity x Sex x three-level diagnosis group: $b = 0.1192$, $SE = 0.2195$, $z = 0.543$, $p = .587$, $OR = 1.127$

```
print_lr_line(right_dx_mod_lr)
```

Right: add diagnosis-group moderation terms: LR chi-square(6) = 5.909, $p = .433$

```
cat_line("\n4C. Separate ANCOVAs testing whether HPC-VS connectivity differs by diagnosis group")
```

4C. Separate ANCOVAs testing whether HPC-VS connectivity differs by diagnosis group

```
cat_line("  Diagnosis groups: No AUD, DSM-IV alcohol abuse, DSM-IV alcohol dependence.")
```

Diagnosis groups: No AUD, DSM-IV alcohol abuse, DSM-IV alcohol dependence.

```

left_conn_reduced <- lm(c_left_hpc_vs ~ Sex + c_ICV, data = analysis_dat)
left_conn_full <- lm(c_left_hpc_vs ~ DxGroup + Sex + c_ICV, data = analysis_dat)
right_conn_reduced <- lm(c_right_hpc_vs ~ Sex + c_ICV, data = analysis_dat)
right_conn_full <- lm(c_right_hpc_vs ~ DxGroup + Sex + c_ICV, data = analysis_dat)
left_conn_ancova <- lm_block_test(left_conn_reduced, left_conn_full, "Left HPC-VS connectivity: add dia
right_conn_ancova <- lm_block_test(right_conn_reduced, right_conn_full, "Right HPC-VS connectivity: add
print_lm_block_line(left_conn_ancova)

```

Left HPC-VS connectivity: add diagnosis group: $F(2, 588) = 0.651$, $p = .522$

```
print_lm_block_line(right_conn_ancova)
```

Right HPC-VS connectivity: add diagnosis group: $F(2, 588) = 0.119$, $p = .888$

```

left_conn_emm <- emmeans::emmeans(left_conn_full, ~ DxGroup)
right_conn_emm <- emmeans::emmeans(right_conn_full, ~ DxGroup)
diagnosis_connectivity_emmeans <- bind_rows(
  as_tibble(as.data.frame(summary(left_conn_emm))) %>%
    mutate(hemisphere = "Left", .before = 1),
  as_tibble(as.data.frame(summary(right_conn_emm))) %>%
    mutate(hemisphere = "Right", .before = 1)
)
diagnosis_connectivity_pairwise <- bind_rows(
  as_tibble(as.data.frame(summary(pairs(left_conn_emm), infer = c(TRUE, TRUE)))) %>%
    mutate(hemisphere = "Left", .before = 1),

```

```

as_tibble(as.data.frame(summary(pairs(right_conn_emm), infer = c(TRUE, TRUE)))) %>%
  mutate(hemisphere = "Right", .before = 1)
)
cat_line("\n Estimated marginal means for z-scored HPC-VS connectivity by diagnosis group")

```

Estimated marginal means for z-scored HPC-VS connectivity by diagnosis group

```
print(diagnosis_connectivity_emmeans)
```

A tibble: 6 x 7

```

hemisphere DxGroup emmean SE df lower.CL upper.CL 1 Left No AUD -0.0152 0.0470 588 -0.108 0.0772 2
Left Abuse 0.106 0.0982 588 -0.0869 0.299 3 Left Dependence 0.0626 0.204 588 -0.338 0.463 4 Right No AUD
0.0109 0.0471 588 -0.0815 0.103 5 Right Abuse -0.0422 0.0983 588 -0.235 0.151 6 Right Dependence -0.00689
0.204 588 -0.407 0.394

```

```

results_primary <- bind_rows(
  left_main_lr,
  right_main_lr,
  left_sex_lr,
  left_int_lr,
  right_sex_lr,
  right_int_lr
)

results_permutation <- bind_rows(
  left_main_perm,
  right_main_perm,
  left_sex_perm,
  left_int_perm,
  right_sex_perm,
  right_int_perm
)

results_dfbetas <- bind_rows(
  left_main_dfb$result,
  right_main_dfb$result,
  left_int_dfb$result,
  right_int_dfb$result
)

results_slopes <- bind_rows(
  left_slopes_full %>% mutate(hemisphere = "Left", .before = 1),
  left_slopes_rms %>% mutate(hemisphere = "Left", .before = 1),
  left_slopes_bkw %>% mutate(hemisphere = "Left", .before = 1),
  right_slopes_full %>% mutate(hemisphere = "Right", .before = 1),
  right_slopes_rms %>% mutate(hemisphere = "Right", .before = 1),
  right_slopes_bkw %>% mutate(hemisphere = "Right", .before = 1)
)

results_age_sensitivity <- bind_rows(

```

```

    left_age_lr,
    right_age_lr,
    left_age_mod_lr,
    right_age_mod_lr
  )

results_diagnosis_moderation <- bind_rows(
  left_dx_mod_lr,
  right_dx_mod_lr
)

results_diagnosis_ancova <- bind_rows(
  left_conn_ancova,
  right_conn_ancova
)

readr::write_csv(results_primary, file.path(OUTPUT_DIR, "primary_lr_tests.csv"))
readr::write_csv(results_permutation, file.path(OUTPUT_DIR, "permutation_lr_tests.csv"))
readr::write_csv(results_dfbetas, file.path(OUTPUT_DIR, "dfbetas_sensitivity_results.csv"))
readr::write_csv(results_slopes, file.path(OUTPUT_DIR, "right_interaction_simple_slopes.csv"))
readr::write_csv(results_slopes, file.path(OUTPUT_DIR, "left_right_interaction_simple_slopes.csv"))
readr::write_csv(results_age_sensitivity, file.path(OUTPUT_DIR, "age_sensitivity_results.csv"))
readr::write_csv(diagnosis_counts, file.path(OUTPUT_DIR, "diagnosis_counts.csv"))
readr::write_csv(results_diagnosis_moderation, file.path(OUTPUT_DIR, "diagnosis_moderation_ordinal_results.csv"))
readr::write_csv(results_diagnosis_ancova, file.path(OUTPUT_DIR, "diagnosis_group_connectivity_ancova_results.csv"))
readr::write_csv(diagnosis_connectivity_emmeans, file.path(OUTPUT_DIR, "diagnosis_group_connectivity_emmeans.csv"))
readr::write_csv(diagnosis_connectivity_pairwise, file.path(OUTPUT_DIR, "diagnosis_group_connectivity_pairwise.csv"))

cat_line("\n", strrep("=", 90))

```

```
cat_line("OUTPUT FILES")
```

OUTPUT FILES

```
cat_line(strrep("=", 90))
```

```
cat_line("Wrote output directory: ", OUTPUT_DIR)
```

Wrote output directory: /Users/tup45568/Downloads/HPC_VC_BingeDrinking/HPC_Ordinal_ManuscriptOrdered_Permutation

```
cat_line("  primary_lr_tests.csv")
```

primary_lr_tests.csv

```
cat_line("  permutation_lr_tests.csv")
```

permutation_lr_tests.csv

```
cat_line(" dfbetas_sensitivity_results.csv")
```

dfbetas_sensitivity_results.csv

```
cat_line(" right_interaction_simple_slopes.csv")
```

right_interaction_simple_slopes.csv

```
cat_line(" left_right_interaction_simple_slopes.csv")
```

left_right_interaction_simple_slopes.csv

```
cat_line(" age_sensitivity_results.csv")
```

age_sensitivity_results.csv

```
cat_line(" diagnosis_counts.csv")
```

diagnosis_counts.csv

```
cat_line(" diagnosis_moderation_ordinal_results.csv")
```

diagnosis_moderation_ordinal_results.csv

```
cat_line(" diagnosis_group_connectivity_ancova_results.csv")
```

diagnosis_group_connectivity_ancova_results.csv

```
cat_line(" diagnosis_group_connectivity_emmeans.csv")
```

diagnosis_group_connectivity_emmeans.csv

```
cat_line(" diagnosis_group_connectivity_pairwise.csv")
```

diagnosis_group_connectivity_pairwise.csv

```
cat_line(" model-specific DFBETAS long/case CSVs")
```

model-specific DFBETAS long/case CSVs

Final Figure 2

This section regenerates the sex-stratified drinking-frequency distribution figure and saves PNG/TIF versions at 300 dpi.

```

options(contrasts = c("contr.sum", "contr.poly"))

suppressPackageStartupMessages({
  library(tidyverse)
  library(rio)
  library(ggplot2)
  library(scales)
})

DATA_PATH <- "/Users/tup45568/Library/CloudStorage/OneDrive-TempleUniversity/0.Experiments/HPC_Drinking

args <- commandArgs(trailingOnly = FALSE)
file_arg <- args[grepl("^--file=", args)]
SCRIPT_DIR <- if (length(file_arg) == 1) dirname(normalizePath(sub("^--file=", "", file_arg))) else getwd()
OUTPUT_DIR <- file.path(SCRIPT_DIR, "Figures")
dir.create(OUTPUT_DIR, recursive = TRUE, showWarnings = FALSE)

HCP_dat <- rio::import(DATA_PATH)

analysis_dat <- HCP_dat %>%
  mutate(
    Subject = as.character(Subject),
    Sex = factor(trimws(Gender), levels = c("F", "M")),
    binge_freq_rc = case_when(
      SSAGA_Alc_12_Frq_5plus == 5 ~ 1L,
      SSAGA_Alc_12_Frq_5plus == 4 ~ 2L,
      SSAGA_Alc_12_Frq_5plus == 3 ~ 3L,
      SSAGA_Alc_12_Frq_5plus %in% c(1L, 2L) ~ 4L,
      TRUE ~ NA_integer_
    ),
    binge_freq_rc_f = factor(
      binge_freq_rc,
      levels = c(1, 2, 3, 4),
      labels = c("Never", "1-11 d/year", "1-3 d/month", "Weekly or greater"),
      ordered = TRUE
    )
  ) %>%
  transmute(
    Subject,
    Sex,
    y = binge_freq_rc_f,
    left_hpc_vs = as.numeric(limbic_Left),
    right_hpc_vs = as.numeric(limbic_Right),
    ICV = as.numeric(FS_IntraCranial_Vol),
    Age = as.numeric(Age_in_Yrs)
  ) %>%
  filter(
    is.finite(left_hpc_vs),
    is.finite(right_hpc_vs),
    is.finite(ICV),
    is.finite(Age),
    !is.na(y),
    !is.na(Sex)
  )

```

```

) %>%
droplevels()

plot_dat_dist <- analysis_dat %>%
  count(Sex, y, name = "n") %>%
  group_by(Sex) %>%
  mutate(
    prop = n / sum(n),
    label = if_else(
      prop < .13,
      paste0(n, " (", percent(prop, accuracy = 1), ")"),
      paste0(n, "\n(", percent(prop, accuracy = 1), ")")
    ),
    text_color = if_else(y == "Weekly or greater", "white", "black")
  ) %>%
  ungroup()

write_csv(plot_dat_dist, file.path(OUTPUT_DIR, "Figure2_DrinkingFrequencyBySex_counts.csv"))

fig_dist <- ggplot(plot_dat_dist, aes(x = Sex, y = prop, fill = y)) +
  geom_col(width = 0.68, color = "white", linewidth = 1.0) +
  geom_text(
    aes(label = label, color = text_color),
    position = position_stack(vjust = 0.5),
    size = 6.0,
    lineheight = 0.88,
    fontface = "bold"
  ) +
  scale_color_identity() +
  scale_y_continuous(
    labels = percent_format(accuracy = 1),
    expand = expansion(mult = c(0, 0.02))
  ) +
  scale_fill_manual(
    values = c(
      "Never" = "#d8eaf6",
      "1-11 d/year" = "#93c5fd",
      "1-3 d/month" = "#3b82f6",
      "Weekly or greater" = "#1e3a8a"
    )
  ) +
  labs(
    x = "Sex",
    y = "Percent within sex",
    fill = "Past-year 5+ drinking frequency",
    title = "Distribution of past-year 5+ drinking\nfrequency by sex"
  ) +
  theme_bw(base_size = 24) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 30, lineheight = 0.94, margin = margin(
    axis.title = element_text(size = 27),
    axis.text = element_text(size = 25, color = "gray15"),
    legend.title = element_text(size = 23),

```

```

    legend.text = element_text(size = 22),
    legend.key.height = unit(0.34, "in"),
    legend.key.width = unit(0.44, "in"),
    legend.position = "right",
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    panel.grid.major.y = element_line(color = "gray88", linewidth = 0.7),
    plot.margin = margin(18, 24, 18, 24)
  )

png_path <- file.path(OUTPUT_DIR, "Figure2_DrinkingFrequencyBySex_LargerFont_300dpi.png")
tif_path <- file.path(OUTPUT_DIR, "Figure2_DrinkingFrequencyBySex_LargerFont_300dpi.tif")

ggsave(png_path, fig_dist, width = 13.5, height = 8.2, units = "in", dpi = 300, bg = "white")
ggsave(tif_path, fig_dist, width = 13.5, height = 8.2, units = "in", dpi = 300, bg = "white", compressi

cat("Analysis N:", nrow(analysis_dat), "\n")

## Analysis N: 593

cat("Sex counts:\n")

## Sex counts:

print(count(analysis_dat, Sex))

##   Sex    n
## 1   F 332
## 2   M 261

cat("Drinking-frequency counts used in Figure 2:\n")

## Drinking-frequency counts used in Figure 2:

print(plot_dat_dist %>% dplyr::select(Sex, y, n, prop))

## # A tibble: 8 x 4
##   Sex    y          n prop
##   <fct> <ord>      <int> <dbl>
## 1 F      Never      143 0.431
## 2 F      1-11 d/year  106 0.319
## 3 F      1-3 d/month   46 0.139
## 4 F      Weekly or greater 37 0.111
## 5 M      Never       66 0.253
## 6 M      1-11 d/year   62 0.238
## 7 M      1-3 d/month   56 0.215
## 8 M      Weekly or greater 77 0.295

```

```
cat(png_path, "\n", sep = "")
```

```
## /Users/tup45568/Downloads/HPC_VC_BingeDrinking/Figures/Figure2_DrinkingFrequencyBySex_LargerFont_300
```

```
cat(tif_path, "\n", sep = "")
```

```
## /Users/tup45568/Downloads/HPC_VC_BingeDrinking/Figures/Figure2_DrinkingFrequencyBySex_LargerFont_300
```

Final Figures 3 And 4

This section regenerates the pooled and sex-stratified HPC-VS ordinal association figures with bootstrap confidence ribbons and saves PNG/TIF versions at 300 dpi.

```
options(contrasts = c("contr.sum", "contr.poly"))
set.seed(23)

suppressPackageStartupMessages({
  library(tidyverse)
  library(MASS)
  library(rio)
  library(emmeans)
  library(ggplot2)
  library(patchwork)
})

DATA_PATH <- "/Users/tup45568/Library/CloudStorage/OneDrive-TempleUniversity/0.Experiments/HPC_Drinking"

args <- commandArgs(trailingOnly = FALSE)
file_arg <- args[grepl("^--file=", args)]
SCRIPT_DIR <- if (length(file_arg) == 1) dirname(normalizePath(sub("^--file=", "", file_arg))) else getwd()
OUTPUT_DIR <- file.path(SCRIPT_DIR, "Figures")
dir.create(OUTPUT_DIR, recursive = TRUE, showWarnings = FALSE)

fit_polr_safe <- function(formula, data) {
  suppressWarnings(
    MASS::polr(formula, data = data, Hess = TRUE, method = "logistic", model = TRUE)
  )
}

boot_expected_ci <- function(data, formula, grid, B = 500, seed = 23) {
  set.seed(seed)
  n <- nrow(data)
  boot_mat <- matrix(NA_real_, nrow = nrow(grid), ncol = B)

  for (b in seq_len(B)) {
    idx <- sample.int(n, size = n, replace = TRUE)
    dat_b <- droplevels(data[idx, , drop = FALSE])

    fit_b <- try(
      suppressWarnings(MASS::polr(formula, data = dat_b, Hess = FALSE, method = "logistic")),

```

```

    silent = TRUE
  )

  if (!inherits(fit_b, "try-error")) {
    vals <- try(expected_from_polr(fit_b, grid), silent = TRUE)
    if (!inherits(vals, "try-error")) {
      boot_mat[, b] <- vals
    }
  }
}

tibble(
  ymin = apply(boot_mat, 1, quantile, probs = 0.025, na.rm = TRUE),
  ymax = apply(boot_mat, 1, quantile, probs = 0.975, na.rm = TRUE)
)

}

expected_from_polr <- function(model, newdata) {
  probs <- as.data.frame(predict(model, newdata = newdata, type = "probs"))
  probs[, "Never"] * 1 +
  probs[, "1-11 d/year"] * 2 +
  probs[, "1-3 d/month"] * 3 +
  probs[, "Weekly or greater"] * 4
}

coef_table_fig <- function(model) {
  ct <- coef(summary(model))
  tibble(
    term = rownames(ct),
    b = ct[, "Value"],
    SE = ct[, "Std. Error"],
    z = ct[, "t value"],
    p = 2 * pnorm(abs(ct[, "t value"]), lower.tail = FALSE)
  )
}

simple_slopes_fig <- function(model, var_name) {
  trends <- emmeans::emtrends(model, specs = ~ Sex, var = var_name, mode = "latent")
  bind_rows(
    as_tibble(as.data.frame(summary(trends, infer = c(TRUE, TRUE)))) %>%
      transmute(
        contrast = as.character(Sex),
        b = !!sym(paste0(var_name, ".trend")),
        p = p.value
      ),
    as_tibble(as.data.frame(summary(pairs(trends), infer = c(TRUE, TRUE)))) %>%
      transmute(
        contrast = as.character(contrast),
        b = estimate,
        p = p.value
      )
  )
}

```

```

fmt_b2 <- function(x) formatC(x, format = "f", digits = 2)

dfbetas_results_path <- file.path(
  SCRIPT_DIR,
  "HPC_Ordinal_ManuscriptOrdered_Permutation_DFBETAS_Output",
  "dfbetas_sensitivity_results.csv"
)
dfbetas_results_for_figures <- if (file.exists(dfbetas_results_path)) {
  readr::read_csv(dfbetas_results_path, show_col_types = FALSE)
} else {
  tibble(model = character(), cutoff_label = character(), p = numeric())
}

slopes_results_path <- file.path(
  SCRIPT_DIR,
  "HPC_Ordinal_ManuscriptOrdered_Permutation_DFBETAS_Output",
  "left_right_interaction_simple_slopes.csv"
)
slopes_results_for_figures <- if (file.exists(slopes_results_path)) {
  readr::read_csv(slopes_results_path, show_col_types = FALSE)
} else {
  tibble(hemisphere = character(), analysis = character(), contrast = character(), p = numeric())
}

if (!"hemisphere" %in% names(slopes_results_for_figures) && nrow(slopes_results_for_figures) > 0) {
  rows_per_hemi <- nrow(slopes_results_for_figures) / 2
  if (rows_per_hemi == floor(rows_per_hemi)) {
    slopes_results_for_figures <- slopes_results_for_figures %>%
      mutate(
        hemisphere = rep(c("Left", "Right"), each = rows_per_hemi),
        .before = 1
      )
  } else {
    slopes_results_for_figures <- slopes_results_for_figures %>%
      mutate(hemisphere = "Right", .before = 1)
  }
}

if (!"analysis" %in% names(slopes_results_for_figures)) {
  slopes_results_for_figures <- slopes_results_for_figures %>%
    mutate(analysis = NA_character_)
}

if (!"contrast" %in% names(slopes_results_for_figures)) {
  slopes_results_for_figures <- slopes_results_for_figures %>%
    mutate(contrast = NA_character_)
}

plus_after_dfbetas <- function(model_name) {
  out <- dfbetas_results_for_figures %>%
    filter(.data$model == model_name, .data$cutoff_label == "max |DFBETAS| > .20")
  nrow(out) > 0 && isTRUE(out$p[1] < 0.05)
}

```

```

sig_marks <- function(p, plus = FALSE) {
  paste0(ifelse(isTRUE(p < 0.05), "*", ""), ifelse(plus, "+", ""))
}

main_effect_label <- function(model, term, model_name) {
  x <- coef_table_fig(model) %>% filter(.data$term == !!term)
  paste0("\U0001D44F = ", fmt_b2(x$b), sig_marks(x$p, FALSE))
}

sex_slope_label <- function(model, var_name, model_name) {
  slopes <- simple_slopes_fig(model, var_name)
  f <- slopes %>% filter(.data$contrast == "F")
  m <- slopes %>% filter(.data$contrast == "M")
  diff <- slopes %>% filter(.data$contrast == "F - M")
  hemisphere <- ifelse(grepl("^left", model_name), "Left", "Right")
  conservative_slope_sig <- function(contrast_label) {
    if (hemisphere != "Right") return(FALSE)
    out <- slopes_results_for_figures %>%
      filter(
        .data$hemisphere == hemisphere,
        .data$analysis == "After max |DFBETAS| > 2/sqrt(N) removal",
        .data$contrast == contrast_label
      )
    nrow(out) > 0 && isTRUE(out$p[1] < 0.05)
  }
  f_plus <- conservative_slope_sig("F")
  m_plus <- conservative_slope_sig("M")
  paste0(
    "F: \U0001D44F = ", fmt_b2(f$b), sig_marks(f$p, f_plus), "\n",
    "M: \U0001D44F = ", fmt_b2(m$b), sig_marks(m$p, m_plus), "\n",
    "F-M: \U0001D44F = ", fmt_b2(diff$b), ifelse(isTRUE(diff$p < 0.05), "*", "")
  )
}

HCP_dat <- rio::import(DATA_PATH)

analysis_dat <- HCP_dat %>%
  mutate(
    Subject = as.character(Subject),
    Sex = factor(trimws(Gender), levels = c("F", "M")),
    binge_freq_rc = case_when(
      SSAGA_Alc_12_Frq_5plus == 5 ~ 1L,
      SSAGA_Alc_12_Frq_5plus == 4 ~ 2L,
      SSAGA_Alc_12_Frq_5plus == 3 ~ 3L,
      SSAGA_Alc_12_Frq_5plus %in% c(1L, 2L) ~ 4L,
      TRUE ~ NA_integer_
    ),
    binge_freq_rc_f = factor(
      binge_freq_rc,
      levels = c(1, 2, 3, 4),
      labels = c("Never", "1-11 d/year", "1-3 d/month", "Weekly or greater"),
      ordered = TRUE
    )
  )

```

```

) %>%
transmute(
  row_id = row_number(),
  Subject,
  Sex,
  y = binge_freq_rc_f,
  left_hpc_vs = as.numeric(limbic_Left),
  right_hpc_vs = as.numeric(limbic_Right),
  ICV = as.numeric(FS_IntraCranial_Vol),
  Age = as.numeric(Age_in_Yrs)
) %>%
filter(
  is.finite(left_hpc_vs),
  is.finite(right_hpc_vs),
  is.finite(ICV),
  is.finite(Age),
  !is.na(y),
  !is.na(Sex)
) %>%
mutate(
  c_left_hpc_vs = as.numeric(scale(left_hpc_vs)),
  c_right_hpc_vs = as.numeric(scale(right_hpc_vs)),
  c_ICV = as.numeric(scale(ICV))
) %>%
droplevels()

left_dat <- analysis_dat %>%
  dplyr::select(row_id, Subject, Sex, y, c_left_hpc_vs, c_ICV) %>%
  droplevels()

right_dat <- analysis_dat %>%
  dplyr::select(row_id, Subject, Sex, y, c_right_hpc_vs, c_ICV) %>%
  droplevels()

left_main <- fit_polr_safe(y ~ c_ICV + c_left_hpc_vs, data = left_dat)
right_main <- fit_polr_safe(y ~ c_ICV + c_right_hpc_vs, data = right_dat)
left_int <- fit_polr_safe(y ~ c_ICV + c_left_hpc_vs * Sex, data = left_dat)
right_int <- fit_polr_safe(y ~ c_ICV + c_right_hpc_vs * Sex, data = right_dat)

left_main_label <- main_effect_label(left_main, "c_left_hpc_vs", "left_main")
right_main_label <- main_effect_label(right_main, "c_right_hpc_vs", "right_main")
left_sex_label <- sex_slope_label(left_int, "c_left_hpc_vs", "left_interaction")
right_sex_label <- sex_slope_label(right_int, "c_right_hpc_vs", "right_interaction")

left_main_grid <- tibble(
  c_ICV = 0,
  c_left_hpc_vs = seq(min(left_dat$c_left_hpc_vs), max(left_dat$c_left_hpc_vs), length.out = 200)
)
left_main_grid$expected_y <- expected_from_polr(left_main, left_main_grid)
left_main_grid <- bind_cols(
  left_main_grid,
  boot_expected_ci(left_dat, y ~ c_ICV + c_left_hpc_vs, left_main_grid, seed = 3101)
)

```

```

right_main_grid <- tibble(
  c_ICV = 0,
  c_right_hpc_vs = seq(min(right_dat$c_right_hpc_vs), max(right_dat$c_right_hpc_vs), length.out = 200)
)
right_main_grid$expected_y <- expected_from_polr(right_main, right_main_grid)
right_main_grid <- bind_cols(
  right_main_grid,
  boot_expected_ci(right_dat, y ~ c_ICV + c_right_hpc_vs, right_main_grid, seed = 3102)
)

left_sex_grid <- expand_grid(
  c_ICV = 0,
  Sex = levels(left_dat$Sex),
  c_left_hpc_vs = seq(min(left_dat$c_left_hpc_vs), max(left_dat$c_left_hpc_vs), length.out = 200)
)
left_sex_grid$expected_y <- expected_from_polr(left_int, left_sex_grid)
left_sex_grid <- bind_cols(
  left_sex_grid,
  boot_expected_ci(left_dat, y ~ c_ICV + c_left_hpc_vs * Sex, left_sex_grid, seed = 3103)
)

right_sex_grid <- expand_grid(
  c_ICV = 0,
  Sex = levels(right_dat$Sex),
  c_right_hpc_vs = seq(min(right_dat$c_right_hpc_vs), max(right_dat$c_right_hpc_vs), length.out = 200)
)
right_sex_grid$expected_y <- expected_from_polr(right_int, right_sex_grid)
right_sex_grid <- bind_cols(
  right_sex_grid,
  boot_expected_ci(right_dat, y ~ c_ICV + c_right_hpc_vs * Sex, right_sex_grid, seed = 3104)
)

y_scale <- scale_y_continuous(
  breaks = 1:4,
  labels = c("Never", "1-11 d/year", "1-3 d/month", "Weekly or greater"),
  limits = c(0.75, 4.25)
)

base_assoc_theme <- theme_bw(base_size = 26) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 36, margin = margin(b = 10)),
    axis.title = element_text(size = 31),
    axis.title.x = element_text(margin = margin(t = 8)),
    axis.title.y = element_text(margin = margin(r = 12)),
    axis.text = element_text(size = 28, color = "gray20"),
    legend.title = element_text(size = 30),
    legend.text = element_text(size = 28),
    legend.key.height = unit(0.28, "in"),
    legend.key.width = unit(0.38, "in"),
    panel.grid.minor = element_blank(),
    panel.grid.major = element_line(color = "gray90", linewidth = 0.7),
    plot.margin = margin(14, 22, 14, 22)
  )

```

```

fig_left_pooled <- ggplot(left_dat, aes(x = c_left_hpc_vs, y = as.numeric(y))) +
  geom_jitter(height = 0.08, width = 0, alpha = 0.38, size = 2.3, color = "black") +
  geom_ribbon(
    data = left_main_grid,
    aes(x = c_left_hpc_vs, y = expected_y, ymin = ymin, ymax = ymax),
    inherit.aes = FALSE,
    alpha = 0.18,
    fill = "#1f5fd2"
  ) +
  geom_line(data = left_main_grid, aes(y = expected_y), linewidth = 1.9, color = "#1f5fd2") +
  annotate("text", x = 4.15, y = 0.88, label = left_main_label, hjust = 1, vjust = 0,
    size = 8.2, fontface = "bold") +
  y_scale +
  coord_cartesian(xlim = c(-2.6, 4.3)) +
  labs(
    x = "HPC-VS connectivity\n(left, z-scored)",
    y = "Past-year 5+ drinking frequency",
    title = "A. Left hemisphere"
  ) +
  base_assoc_theme

fig_right_pooled <- ggplot(right_dat, aes(x = c_right_hpc_vs, y = as.numeric(y))) +
  geom_jitter(height = 0.08, width = 0, alpha = 0.38, size = 2.3, color = "black") +
  geom_ribbon(
    data = right_main_grid,
    aes(x = c_right_hpc_vs, y = expected_y, ymin = ymin, ymax = ymax),
    inherit.aes = FALSE,
    alpha = 0.18,
    fill = "#1f5fd2"
  ) +
  geom_line(data = right_main_grid, aes(y = expected_y), linewidth = 1.9, color = "#1f5fd2") +
  annotate("text", x = 4.15, y = 0.88, label = right_main_label, hjust = 1, vjust = 0,
    size = 8.2, fontface = "bold") +
  y_scale +
  coord_cartesian(xlim = c(-2.6, 4.3)) +
  labs(
    x = "HPC-VS connectivity\n(right, z-scored)",
    y = "Past-year 5+ drinking frequency",
    title = "B. Right hemisphere"
  ) +
  base_assoc_theme +
  theme(
    axis.title.y = element_blank(),
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank()
  )

fig_assoc_pooled <- fig_left_pooled + fig_right_pooled + plot_layout(widths = c(1, 1))

fig_left_by_sex <- ggplot(left_dat, aes(x = c_left_hpc_vs, y = as.numeric(y), color = Sex)) +
  geom_jitter(height = 0.08, width = 0, alpha = 0.46, size = 2.3) +
  geom_ribbon(
    data = left_sex_grid,

```

```

aes(x = c_left_hpc_vs, y = expected_y, ymin = ymin, ymax = ymax, fill = Sex),
inherit.aes = FALSE,
alpha = 0.14,
color = NA
) +
geom_line(data = left_sex_grid, aes(y = expected_y), linewidth = 1.9) +
annotate("text", x = 4.15, y = 0.88, label = left_sex_label, hjust = 1, vjust = 0,
         size = 7.2, fontface = "bold", lineheight = 0.95, color = "black") +
scale_color_manual(values = c("F" = "#1f5fd2", "M" = "#f39c12")) +
scale_fill_manual(values = c("F" = "#1f5fd2", "M" = "#f39c12"), guide = "none") +
y_scale +
coord_cartesian(xlim = c(-2.6, 4.3)) +
labs(
  x = "HPC-VS connectivity\n(left, z-scored)",
  y = "Past-year 5+ drinking frequency",
  title = "A. Left hemisphere",
  color = "Sex"
) +
base_assoc_theme

fig_right_by_sex <- ggplot(right_dat, aes(x = c_right_hpc_vs, y = as.numeric(y), color = Sex)) +
geom_jitter(height = 0.08, width = 0, alpha = 0.46, size = 2.3) +
geom_ribbon(
  data = right_sex_grid,
  aes(x = c_right_hpc_vs, y = expected_y, ymin = ymin, ymax = ymax, fill = Sex),
  inherit.aes = FALSE,
  alpha = 0.14,
  color = NA
) +
geom_line(data = right_sex_grid, aes(y = expected_y), linewidth = 1.9) +
annotate("text", x = 4.15, y = 0.88, label = right_sex_label, hjust = 1, vjust = 0,
         size = 7.2, fontface = "bold", lineheight = 0.95, color = "black") +
scale_color_manual(values = c("F" = "#1f5fd2", "M" = "#f39c12")) +
scale_fill_manual(values = c("F" = "#1f5fd2", "M" = "#f39c12"), guide = "none") +
y_scale +
coord_cartesian(xlim = c(-2.6, 4.3)) +
labs(
  x = "HPC-VS connectivity\n(right, z-scored)",
  y = "Past-year 5+ drinking frequency",
  title = "B. Right hemisphere",
  color = "Sex"
) +
base_assoc_theme +
theme(
  axis.title.y = element_blank(),
  axis.text.y = element_blank(),
  axis.ticks.y = element_blank()
)

fig_assoc_by_sex <- fig_left_by_sex + fig_right_by_sex + plot_layout(widths = c(1, 1), guides = "collect")
theme(legend.position = "right")

save_figure <- function(plot, basename, width = 18.5, height = 10.2) {

```

```

png_path <- file.path(OUTPUT_DIR, paste0(basename, ".png"))
tif_path <- file.path(OUTPUT_DIR, paste0(basename, ".tif"))

ggsave(png_path, plot = plot, width = width, height = height, units = "in", dpi = 300, bg = "white")
ggsave(tif_path, plot = plot, width = width, height = height, units = "in", dpi = 300, bg = "white",
cat(png_path, "\n", sep = "")
cat(tif_path, "\n", sep = "")
}

save_figure(fig_assoc_pooled, "Figure3_HPCVS_Ordinal_Association_Pooled_Annotated_300dpi")

## /Users/tup45568/Downloads/HPC_VC_BingeDrinking/Figures/Figure3_HPCVS_Ordinal_Association_Pooled_Annotated_300dpi.png
## /Users/tup45568/Downloads/HPC_VC_BingeDrinking/Figures/Figure3_HPCVS_Ordinal_Association_Pooled_Annotated_300dpi.tif

save_figure(fig_assoc_by_sex, "Figure4_HPCVS_Ordinal_Association_BySex_Annotated_300dpi")

## /Users/tup45568/Downloads/HPC_VC_BingeDrinking/Figures/Figure4_HPCVS_Ordinal_Association_BySex_Annotated_300dpi.png
## /Users/tup45568/Downloads/HPC_VC_BingeDrinking/Figures/Figure4_HPCVS_Ordinal_Association_BySex_Annotated_300dpi.tif

```

Final Figure 5

This section creates the diagnosis-group HPC-VS connectivity figure reported in the manuscript.

```

set.seed(23)

suppressPackageStartupMessages({
  library(tidyverse)
  library(rio)
  library(ggplot2)
  library(patchwork)
})

DATA_PATH <- "/Users/tup45568/Library/CloudStorage/OneDrive-TempleUniversity/0.Experiments/HPC_Drinking/Connectivity/Figure5_HPCVS_Ordinal_Association_Pooled_Annotated_300dpi.png"

args <- commandArgs(trailingOnly = FALSE)
file_arg <- args[grepl("^--file=", args)]
SCRIPT_DIR <- if (length(file_arg) == 1) dirname(normalizePath(sub("^--file=", "", file_arg))) else getwd()
OUTPUT_DIR <- file.path(SCRIPT_DIR, "Figures")
dir.create(OUTPUT_DIR, recursive = TRUE, showWarnings = FALSE)

fmt_p_fig <- function(p) {
  ifelse(p < 0.001, "< .001", paste0("= ", sub("^0", "", formatC(p, format = "f", digits = 3))))
}

HCP_dat <- rio::import(DATA_PATH)

diagnosis_plot_dat <- HCP_dat %>%
  mutate(
    Subject = as.character(Subject),
    Sex = factor(trimws(Gender), levels = c("F", "M")),

```

```

alcohol_abuse_dx = case_when(
  SSAGA_Alc_D4_Ab_Dx == 5 ~ TRUE,
  SSAGA_Alc_D4_Ab_Dx == 1 ~ FALSE,
  TRUE ~ NA
),
alcohol_dependence_dx = case_when(
  SSAGA_Alc_D4_Dp_Dx == 5 ~ TRUE,
  SSAGA_Alc_D4_Dp_Dx == 1 ~ FALSE,
  TRUE ~ NA
),
DxGroup = factor(
  case_when(
    alcohol_dependence_dx ~ "Dependence",
    alcohol_abuse_dx ~ "Abuse",
    alcohol_abuse_dx == FALSE & alcohol_dependence_dx == FALSE ~ "No AUD",
    TRUE ~ NA_character_
  ),
  levels = c("No AUD", "Abuse", "Dependence")
)
) %>%
transmute(
  Subject,
  Sex,
  DxGroup,
  left_hpc_vs = as.numeric(limbic_Left),
  right_hpc_vs = as.numeric(limbic_Right),
  ICV = as.numeric(FS_IntraCranial_Vol)
) %>%
filter(
  is.finite(left_hpc_vs),
  is.finite(right_hpc_vs),
  is.finite(ICV),
  !is.na(Sex),
  !is.na(DxGroup)
) %>%
mutate(
  c_left_hpc_vs = as.numeric(scale(left_hpc_vs)),
  c_right_hpc_vs = as.numeric(scale(right_hpc_vs)),
  c_ICV = as.numeric(scale(ICV))
) %>%
droplevels()

left_conn_reduced <- lm(c_left_hpc_vs ~ Sex + c_ICV, data = diagnosis_plot_dat)
left_conn_full <- lm(c_left_hpc_vs ~ DxGroup + Sex + c_ICV, data = diagnosis_plot_dat)
right_conn_reduced <- lm(c_right_hpc_vs ~ Sex + c_ICV, data = diagnosis_plot_dat)
right_conn_full <- lm(c_right_hpc_vs ~ DxGroup + Sex + c_ICV, data = diagnosis_plot_dat)

left_p <- anova(left_conn_reduced, left_conn_full)$`Pr(>F)`[2]
right_p <- anova(right_conn_reduced, right_conn_full)$`Pr(>F)`[2]

diagnosis_long <- diagnosis_plot_dat %>%
  dplyr::select(Subject, DxGroup, c_left_hpc_vs, c_right_hpc_vs) %>%
  pivot_longer(

```

```

cols = c(c_left_hpc_vs, c_right_hpc_vs),
names_to = "hemisphere",
values_to = "connectivity"
) %>%
mutate(
  hemisphere = factor(
    hemisphere,
    levels = c("c_left_hpc_vs", "c_right_hpc_vs"),
    labels = c(
      paste0("A. Left hemisphere\nANCOVA p ", fmt_p_fig(left_p)),
      paste0("B. Right hemisphere\nANCOVA p ", fmt_p_fig(right_p))
    )
  )
)

fig_diagnosis <- ggplot(diagnosis_long, aes(x = DxGroup, y = connectivity, fill = DxGroup)) +
  geom_violin(width = 0.92, alpha = 0.36, color = NA, trim = FALSE) +
  geom_boxplot(width = 0.18, outlier.shape = NA, alpha = 0.75, color = "gray20", linewidth = 0.8) +
  geom_jitter(width = 0.08, height = 0, alpha = 0.42, size = 2.1, color = "gray15") +
  stat_summary(fun = mean, geom = "point", shape = 23, size = 4.6, fill = "white", color = "black") +
  facet_wrap(~ hemisphere, nrow = 1) +
  scale_fill_manual(values = c("No AUD" = "#89e6a8", "Abuse" = "#87bff5", "Dependence" = "#f6b737"), gu
  labs(
    title = "HPC-VS Connectivity by Alcohol Diagnosis Group",
    x = "Lifetime DSM-IV alcohol diagnosis group",
    y = "HPC-VS connectivity (z-scored)"
  ) +
  theme_bw(base_size = 24) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 34, margin = margin(b = 10)),
    strip.background = element_rect(fill = "gray85", color = "gray20", linewidth = 0.9),
    strip.text = element_text(face = "bold", size = 25, lineheight = 0.95),
    axis.title = element_text(size = 25),
    axis.text = element_text(size = 22, color = "gray20"),
    panel.grid.minor = element_blank(),
    panel.grid.major = element_line(color = "gray90", linewidth = 0.7),
    plot.margin = margin(14, 22, 14, 22)
  )

png_path <- file.path(OUTPUT_DIR, "Figure_DiagnosisGroup_HPCVS_Connectivity_300dpi.png")
tif_path <- file.path(OUTPUT_DIR, "Figure_DiagnosisGroup_HPCVS_Connectivity_300dpi.tif")
ggsave(png_path, plot = fig_diagnosis, width = 15.8, height = 8.8, units = "in", dpi = 300, bg = "white")
ggsave(tif_path, plot = fig_diagnosis, width = 15.8, height = 8.8, units = "in", dpi = 300, bg = "white")
cat(png_path, "\n", sep = "")

## /Users/tup45568/Downloads/HPC_VC_BingeDrinking/Figures/Figure_DiagnosisGroup_HPCVS_Connectivity_300dpi.png

cat(tif_path, "\n", sep = "")

## /Users/tup45568/Downloads/HPC_VC_BingeDrinking/Figures/Figure_DiagnosisGroup_HPCVS_Connectivity_300dpi.tif

```